

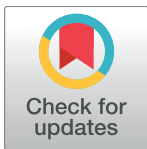
RESEARCH ARTICLE

The effect of social group size on feather corticosterone in the co-operatively breeding Smooth-billed Ani (*Crotophaga ani*): An assay validation and analysis of extreme social living

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Abstract

Living closely with others can provide a myriad of fitness benefits, from shared territory defense to co-operative resource acquisition. Costs of social aggregation are not absent, however, and likely influence optimal and observed groups' sizes in a social species. Here, we explored optimal group size in a joint-nesting cuckoo species (the Smooth-billed Ani, *Crotophaga ani*) using endocrine markers of stress physiology (corticosterone, or CORT). Smooth-billed Anis exhibit intense reproductive competition that is exacerbated in atypically large groups. We therefore hypothesized that intra-group competition (measured by social group size) mediates the desirability and physiological cost of social group membership in this species. To test this hypothesis, we captured 47 adult Smooth-billed Anis (31 males, 16 females) during the breeding seasons of 2012–2014 in south-western Puerto Rico, and documented social group sizes. Tail feathers were sampled and used to quantify CORT (pg/mg) in enzyme-linked immunosorbent assays (ELISAs) ($n = 50$). Our analyses show significant differences in feather-CORT of adults between categorical group sizes, with individuals from atypically large social groups ($\geq x + 1SD$) having highest mean concentrations (33.319 pg/mg), and individuals from atypically small social groups ($\leq x - 1SD$) having lowest mean concentrations (8.969 pg/mg). Whether reproductive competition or effort is responsible for elevated CORT in atypically large social groups, however, remains unclear. Our results suggest that living in atypically large groups is physiologically expensive and may represent an evolutionarily unstable strategy. To our knowledge, this is the first study to explore a correlation between stress physiology and group size in a joint-nesting species.

Introduction

Living in groups can provide numerous fitness benefits, either directly (through shared territory defense, predator defense, resource acquisition, and parental effort) (for example, [1–8]) or indirectly (through dilution of predation risk and inclusive fitness benefits) [9, 10]. Costs of